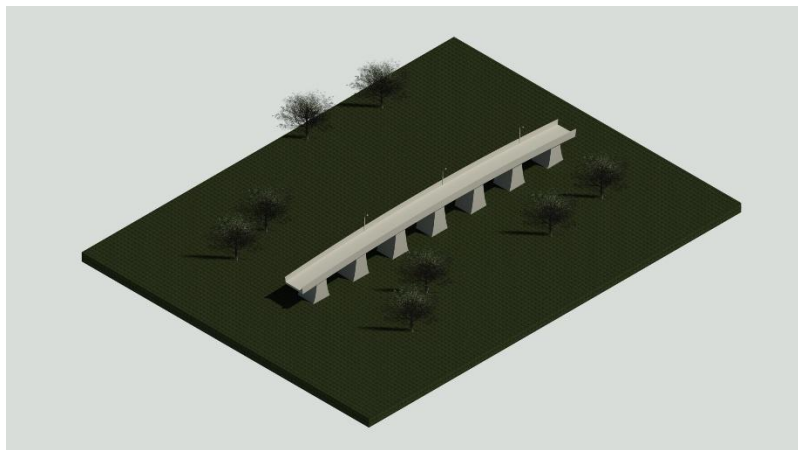


Project Report

Parametric Pedestrian Bridge



Advanced Building Information Modelling
Summer-Term 2025

Date

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List of Abbreviations

Abbreviation	Explanation
BIM	Building Information Modelling
IFC	Industry Foundation Classes
4D	3D Geometry + Time (Schedule Integration)
Revit	Autodesk Revit (BIM software)
RFA	Revit Family File (.rfa)
RVT	Revit Project File (.rvt)
Dynamo	Visual Programming Extension of Revit
BPMN	Business Process Model and Notation

1. Motivation

a. Initial idea

When thinking of an idea for the project, many ideas were discussed like creating building models. We understood that Bridges are relatively hard to model in REVIT. So, to gain experience through the project, a parametric bridge with the BIM process was decided.

The initial idea was to make a bridge with modular segments or arcs which can adapt to user inputs. The usage of parametric family and Dynamo will help to reduce time for modelling.^{[1][2]} Basic aim was to create a standard family like beam, column, etc. with only dimensions and the bridge path as input. The width, height, and length were user inputs.

b. Feedback

While the idea was accepted at the idea presentation. Feedback of implementing a curve in the bridge was given.

2. Revit Model

a. Parametric family

The very first issue began with the Idea Paper itself. Since, as mentioned above, modelling a Bridge in REVIT is more challenging than a building structure due to absence of proper standard families. We came up with a Generic model Adaptive family was used to create a bridge element consisting of the bridge deck and the pillar/pier. The pillar is the sub family (also adaptive). The main aim of using adaptive family is that the model has adaptive points, and all the geometry is in reference to those points.^[1] This was the breakthrough for making a bridge with a curve.^[6]

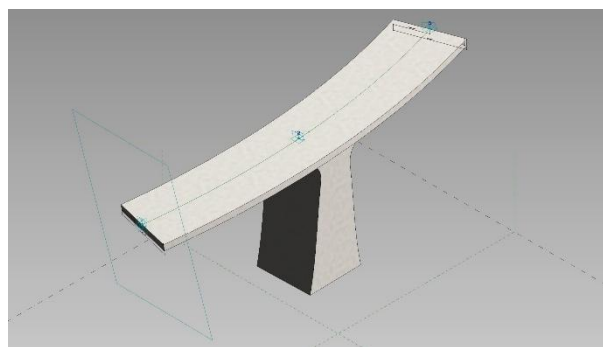


Figure 1: Bridge element (parametric family)

The deck width, pillar height and deck thickness are the parametric dimensions, which can be changed. Although the pillar height is automated by Dynamo, other dimensions can be put as per the user requirement.

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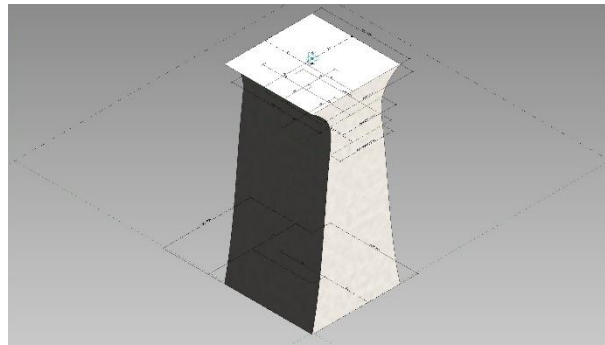


Figure 2: Pillar (sub-parametric family)

Parametric parameter	Family
width	Bridge element
thickness	Bridge element
Depth	Pillar

Table 1: List of parametric dimensions

b. Custom families

Two more custom families of Railing and streetlamp are created to use on the bridge.

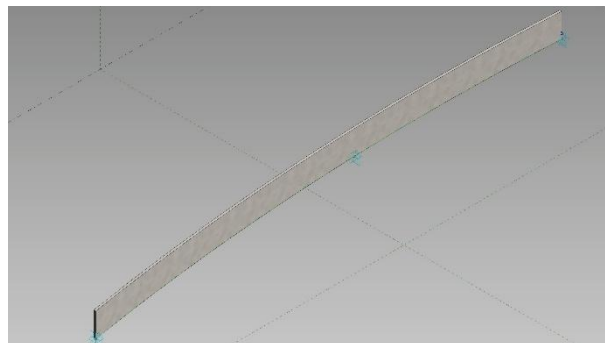


Figure 3: Railing

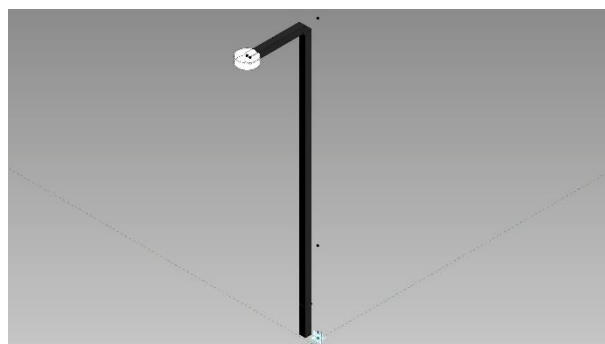


Figure 4: Streetlamp

5. BPMN process map

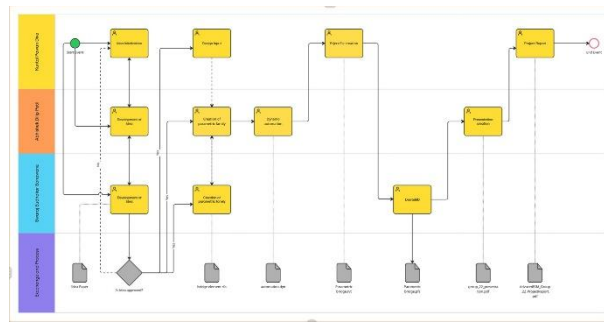


Figure 7: BPMN process map^[7]

All the team members collectively worked on the parametric family which was the main part of the project.

-Kuntal Pawan Dive had the initial idea. Moreover, the final Revit project file was created by importing the families and adding the surrounding. Lastly, the final report and the elevator pitch was done.

-Abhishek Dilip Patil created the Dynamo automation script and the presentation pdf.

-Swaraj Sudhakar Sonawane had major role in creating the Idea paper and the interfacing with DECITEmd and the 4D simulation.

All the individual work done by the team members were done independently to each other while also simultaneously helping with other's tasks. Reason being having varying schedules throughout the semester. The flow of the work went well when it came to stitch together all the aspects of the project in the final stages.

Each team member's contribution played a crucial role in completion of the project. The process flow can be seen in the above Figure 7: BPMN process map.

6. Conclusion

The project demonstrates the potential of BIM to create adaptable and efficient modular infrastructure through the parametric bridge. Instead of using conventional grid-based or linear spacing for pattern creation, we developed a custom pattern using a spline, allowing for greater flexibility in layout. Additionally, we utilized adaptive families to introduce both elevation changes and curvature into the bridge design—an approach that differed significantly from the standard methods used in the exercises.

The Dynamo and DECITEmd were successfully implemented along with REVIT being the base for modelling. Last but not least, the Teamwork was monumental throughout the project. A better understanding of families in REVIT has been gained through the project.^[3]

7. References

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- [4] **Solibri, 2022.** *IFC File Structure and OpenBIM Interoperability*. [online] Available at: <https://www.solibri.com/resources/ifc-guide>.
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- [7] **Miro, 2025.** *Online Collaborative Whiteboard Platform*. [online] Available at: <https://miro.com/>.
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Declaration

I hereby certify that I/we have prepared this report independently and without outside help, and that I/we have marked all passages taken verbatim from other authors, as well as those closely based on the ideas of other authors, and that I/we have cited the sources.

The thesis has not yet been published in the same or a similar form and has not yet been submitted to an examination authority.

Weimar, 16.07.2025

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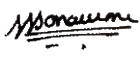
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